DARYA TAGHVA

Artificial Intelligent Engineer

Chertsey, England, United Kingdom | +44 7863653048 | dtaghva26@gmail.com

LinkedIn | ***GitHub***

PROFILE

- AI Engineer specializing in architecting scalable, production-grade RAG and machine learning pipelines. Proven track record of bridging the gap between cutting-edge LLM research and reliable, user-facing applications to optimize process efficiency and data accuracy.

EMPLOYMENT HISTORY

Code Sensei, CodeNinjas

Jan 2026 — Now

- Delivered high-impact coding instruction in JavaScript, Lua, and C#, accelerating student project completion rates by 40% through hands-on technical mentorship.
- Simplified complex syntax (e.g., conditional logic, variables) into intuitive, gamified modules, improving student retention and understanding for non-technical learners.
- Institutionalized systematic debugging strategies, reducing session downtime by 25% by teaching students root-cause analysis and script organization.

Django Internship, Quera

Jul 2023 — Dec 2023

- Engineered a robust, role-based university simulation platform using Django, reducing administrative processing time by 35% through automated grading and registration workflows.
- Spearheaded the development of 50+ RESTful APIs, facilitating seamless integration between core service features; maintained build stability through rigorous code reviews and version control.
- Led development sprints for a 5-person team, slashing ticket turnaround time by 20% through efficient backlog grooming and milestone tracking.

EDUCATION

MSc in Artificial Intelligence, University of Surrey

Sep 2024 — Sep 2025

- Graduated with Merit.
- Dissertation: Sleep audio classification using spectrogram features (Comparison of Mel and CQT spectrograms), deployed on web using Flask.

BSc in Computer Engineering, Shahid Beheshti University

Sep 2019 — Jun 2024

- Dissertation: Development and Deployment of a system for making no-code applications using MERN tech stack.

SKILLS

- **Core Languages:** Python, TypeScript
- **Cloud & MLOps:** Docker, AWS
- **Specialized Domains:** Computer Vision, Signal Processing, NLP
- **ML Ecosystems:** PyTorch, TensorFlow, Skorch
- **AI Models:** HuggingFace
- **ML Governance:** MLflow, Docker, AWS
- **Storage:** MySQL, SQLite, MongoDB
- **Server Architectures:** Flask, FastAPI, Express.js

PROJECTS

Traditional-ML-algorithms

- Built interactive ML visualization dashboard using React 19, TypeScript, D3.js, and Tailwind CSS to explore scikit-learn algorithms and hyperparameter tuning with a force-directed graph interface.
- Engineered bias-variance tradeoff visualization with interactive parameter exploration and algorithm comparison game mode to make machine learning concepts intuitive and educational.
- Integrated Gemini API for intelligent hyperparameter tuning suggestions, combining visualization with AI-powered recommendations for model optimization.

RAG Blog creation system using CrewAI and NLP pipelines

- Multi-Agent content generation pipelines.
- Utilized and highly optimized for UK educational pipelines (Year 1 to Year 6).
- Built a Gradio-based interface for content generation, integrating NLP pipelines to automate blog drafting.

Mental Health index prediction using screen time indexing

Jun 2025 — Aug 2025

- Developed and implemented machine learning models for detecting mental health indices with over 90% accuracy.
- Performed Exploratory Data Analysis (EDA) on screen time data.

Object detection using YOLO model

Mar 2025 — Jun 2025

- Improved YOLO model loss function by integrating Varifocal loss for airplane detection.
- Applied advanced data augmentations using the Albumentations package.

Covid detection Using Chest X-Ray

Feb 2025 — May 2025

- Built a hybrid model using Chest X-Ray images and tabular data to detect COVID-19 with 94% accuracy.

Alzheimer’s detection

Feb 2025 — May 2025

- Designed a machine learning system (Logistic Regression, Gradient Boosting) using ELSA dataset for Dementia detection.

Tourism management system development

Dec 2022 — Jun 2023

- Applied OOP principles to build a reusable and maintainable tourism management application.

Hive AI agent development

Oct 2020 — Dec 2020

- Independently built an AI agent using MiniMax algorithm (depth 7) for the Hive board game.
- Designed a heuristic loss function improving decision-making by 30%.

COURSES

- **Capstone project Deep learning** (May 2026 — Jun 2026)
- **Foundations of Project Management** (Apr 2026 — May 2026)
- **Advanced RAG with Vector Databases and Retrievers** (Mar 2026 — Apr 2026)
- **Agentic AI with LangChain and LangGraph** (Mar 2026 — Apr 2026)
- **Reinforcement Fine tuning, deeplearning.ai** (Oct 2025 — Nov 2025)
- **ChatGPT Prompt Engineering for Developers, deeplearning.ai** (Sep 2025 — Oct 2025)
- **Complete Data science, Machine Learning,DL,NLP Bootcamp, Udemy** (Sep 2025 — Oct 2025)
- **Knowledge Graph for AI Agent API discovery, deeplearning.ai** (Aug 2025 — Sep 2025)